Bohr–Jessen for $\mathrm{U}(N)$

Theorem 1. *Let U_N be Haar distributed on the unitary group $\mathrm{U}(N)$. Fix $u \in \mathbb{C}$ with $|u| < 1$. Then*

$$\det(I - uU_N) \xrightarrow[N\to\infty]{d} \exp\left(-\sum_{k\geq 1} \frac{N_k u^k}{\sqrt{k}} \right),$$

where $(N_k)_{k\geq 1}$ are independent standard complex Gaussian random variables.

Proof. Let $e^{i\theta_1}, \ldots, e^{i\theta_N}$ be the eigenvalues of U_N. Since $|u| < 1$, the power series for $\log(1-z)$ is absolutely convergent at every $z = ue^{i\theta_j}$. Thus

$$\log\det(I - uU_N) = \sum_{j=1}^{N} \log(1 - ue^{i\theta_j}) = -\sum_{k\geq 1} \frac{u^k}{k} \mathrm{Tr}(U_N^k).$$

Therefore, by the continuous mapping theorem it is enough to prove that

$$\sum_{k\geq 1} \frac{u^k}{k} \mathrm{Tr}(U_N^k) \xrightarrow[N\to\infty]{d} \sum_{k\geq 1} \frac{u^k}{\sqrt{k}} N_k. \tag{1}$$

We use the Diaconis–Shahshahani theorem: for every fixed $m \geq 1$,

$$\left(\frac{\mathrm{Tr}(U_N)}{\sqrt{1}}, \frac{\mathrm{Tr}(U_N^2)}{\sqrt{2}}, \ldots, \frac{\mathrm{Tr}(U_N^m)}{\sqrt{m}} \right) \xrightarrow[N\to\infty]{d} (N_1, \ldots, N_m),$$

where $N_1, \ldots, N_m$ are independent standard complex Gaussians. For fixed m, define

$$S_{N,m} := \sum_{k=1}^{m} \frac{u^k}{k} \mathrm{Tr}(U_N^k).$$

By Diaconis–Shahshahani and the continuous mapping theorem,

$$S_{N,m} = \sum_{k=1}^{m} \frac{u^k}{\sqrt{k}} \frac{\mathrm{Tr}(U_N^k)}{\sqrt{k}} \xrightarrow[N\to\infty]{d} S_m := \sum_{k=1}^{m} \frac{u^k}{\sqrt{k}} N_k.$$

It remains to pass from finite sums to infinite sums. Let

$$R_{N,m} := \sum_{k>m} \frac{u^k}{k} \mathrm{Tr}(U_N^k).$$

For Haar unitary matrices one has the covariance identity (due to Dyson)

$$\mathbb{E}\Big[\mathrm{Tr}(U_N^j) \overline{\mathrm{Tr}(U_N^k)} \Big] = \delta_{jk} \min(j, N).$$

Consequently,

$$\mathbb{E}|R_{N,m}|^2 = \sum_{k>m} \frac{|u|^{2k}}{k^2} \min(k, N) \le \sum_{k>m} \frac{|u|^{2k}}{k}.$$

Since $|u| < 1$, the right-hand side tends to 0 as $m \to \infty$, uniformly in N. Hence $R_{N,m} \to 0$ in L^2, uniformly in N, as $m \to \infty$. Similarly, if

$$R_m := \sum_{k>m} \frac{u^k}{\sqrt{k}} N_k,$$

then independence gives

$$\mathbb{E}|R_m|^2 = \sum_{k>m} \frac{|u|^{2k}}{k} \longrightarrow 0.$$

Thus $S_m \to S$ in L^2, where

$$S := \sum_{k\ge1} \frac{u^k}{\sqrt{k}} N_k.$$

Combining $S_{N,m} \xrightarrow[N\to\infty]{d} S_m$ with the uniform L^2-control of the tails yields (1). In more detail, since $S_{N,m} \xrightarrow[N\to\infty]{d} S_m$, $S_m \to S$ in L^2 (and in particular in probability), $\sup_N \mathbb{P}(|R_{N,m}| > \varepsilon) \to 0$ for every $\varepsilon > 0$ and $R_m \xrightarrow[m\to\infty]{p} 0$, we obtain $S_{N,\infty} \xrightarrow[N\to\infty]{d} S$ where $S_{N,\infty} := \sum_{k=1}^{\infty} \frac{u^k}{k} \mathrm{Tr}(U_N^k)$. $\square$